



ELECT MAINT- MAIN PLANT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Date:

12/09/2023(10:49:47)

Subject:Revision of LMI LMI/OGN/OPS/ELEC/006 HAND HELD
INSULATED TOOLS FOR LIVE WORKING UP TO 1000V

LMIIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes are envisaged by COS. Only periodic reviewing done after two years.

No changes are done in the document.

May please approve.

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

KRISHNA BHEESETTY - 009044 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
,Simhadri

(BKRISHNA@NTPC.CO.IN, 9493546044)

On:12/09/2023(10:49:47) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:2

Submitted by:

बाला बालाजी आकुला - उप महाप्रबंधक

BALA BALAJI AKULA - 009145 (DY. GENERAL MANAGER ,ELECT MAINT)
(ABALABALAJI@NTPC.CO.IN, 8332959245) ,Simhadri

On:12/09/2023(10:50:18) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:3

Submitted by:

मेडा नाग राजा - उप महाप्रबंधक

Meda Naga Raja - 008404 (DY. GENERAL MANAGER ,ELECT MAINT)
(MEDANAGARAJA@NTPC.CO.IN, 9437578452) ,Simhadri

On:12/09/2023(15:56:42) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:4

Submitted by:

सतीश कुमार सिंह - उप महाप्रबंधक

Satish Kumar Singh - 008418 (DY. GENERAL MANAGER ,ELECT MAINT)
(SATISHKUMARSINGH@NTPC.CO.IN, 8004940879) ,Simhadri

On:12/09/2023(16:02:25) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:5

Reviewed

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

Makala Nageswara Rao - 007521 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(MAKALANAGESWAR@NTPC.CO.IN, 8331016103) ,Simhadri

On:14/09/2023(18:15:59) **Ref:**SMTTPP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:6

Township EMD may pl see and Vet

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:15/09/2023(17:36:15) **Ref:**SMTTPP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:7

Reviewed

Submitted by:

सुर्या प्रकाश राव पसुपुरेड्डी - वरिष्ठ प्रबंधक

SURYA PRAKASH RAO PASUPUREDDY - 102528 (SENIOR MANAGER ,T/S ELECT MAINT)
(SURYAPRAKASHP@NTPC.CO.IN, 9899678148) ,Simhadri

On:25/09/2023(09:48:36) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:8

LMI reviewed and revised check list incorporated REVISION - 2 , June-2023.

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:25/09/2023(10:22:57) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:9

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:25/09/2023(10:39:10) **Ref:**SMTTP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:10

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)

(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:25/09/2023(11:06:15) **Ref:**SMTPP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:11

LMI revision reviewed and the same may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)

(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:09/10/2023(18:36:11) **Ref:**SMTPP/ELECT MAINT-/2023-24/EMDLMIHH/475991

Note No:12

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)

(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:10/10/2023(10:29:23) **Ref:**SMTPP/ELECT MAINT-/2023-24/EMDLMIHH/475991

NTPC Limited
SIMHADRI THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION
LMI/OGN/OPS/ELEC/006

Issue No: 1, Rev: 2 Date: June 2023

**HAND HELD INSULATED TOOLS FOR LIVE
WORKING UP TO 1000V**

Approved for
Implementation by.....
Group General Manager (Simhadri)

Date:

NTPC Limited

SIMHADRI THERMAL POWER STATION

ELECTRICAL MAINTENANCE DEPARTMENT

LMI	HAND HELD INSULATED TOOLS FOR LIVE WORKING UP TO 1000V		
LMI No	LMI/OGN/OPS/ELEC/006		
ISSUE No	1	REVISION: 2	DATE: June 2023

	DESIGNATION	SIGNATURE & NAME
Prepared by	DGM (EM)	
Checked by	AGM (EM)	
Reviewed by	AGM (EM) I/C	
Reviewed by	GM (Maint)	
Reviewed by	GM(O&M)	

HAND HELD INSULATED TOOLS FOR LIVE WORKING UP TO 1000V

CONTENTS

		PAGE NO
1.0	OBJECTIVE	4
2.0	SCOPE	4
3.0	REFERENCE DOCUMENT	4
4.0	SUPERSEDED DOCUMENTS	4
5.0	REVISION DETAILS	5
6.0	THE SPECIFIC INSTRUCTIONS TO BE FOLLOWED	5
7.0	METHODOLOGY OF IMPLEMENTATION	6
8.0	THE RESPONSIBILITY CENTERS FOR IMPLEMENTATION	8
9.0	THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS	8
10.0	REVIEW	8
11.0	ANNEXURE	9-11

HAND HELD INSULATED TOOLS FOR LIVE WORKING UP TO 1000V

1.0 OBJECTIVE:

There are 3 forms of insulation that apply to the insulated hand Tools.

- i) Partially insulated Tools: tools in which comparatively large areas of bare metal are left the exposed, e.g., pliers and other scissors action tools in which the handles are insulated but the remainder is not.
- ii) Fully Insulated Tools: tools in which the exposed uninsulated metal is very limited, e.g., interior haws of ring or open ended spanners, tips of screwdrivers, etc.
- iii) All Insulated Tools: tools made entirely of insulating materials.

This latter type is use for very specific application and thus does not tend to be on general use i.e. **3.3 KV, 11 KV etc.** Insulated tools for use in live working should not be confused with general engineering hand tools which may have handles constructed of insulating materials for comfort or mechanical reasons, e.g. increase diameter of handle yet keep weight low or absorb mechanical shock.

2.0 SCOPE:

This LMI covers all the aspects involved in procurement, storage, control & issue, usage, inspection and testing details of hand held insulated tools meant for live working up to 1000v

3.0 REFERENCE DOCUMENT:

Operation Guide note COS-ISO-00-OGN/OPS/ELEC/006 Rev. No.:2
May' 2021

4.0 SUPERSEDED DOCUMENT:

NIL

5.0 REVISION DETAILS:

LMI revised in line with revised COS-ISO-00-OGN/OPS/ELEC/006 in May 2021

- (a) Mention of the applicable standard at clause no.6.1 at page – 5*
- (b) Mention for preparation of a policy by station for control and issue of Hand held tools in clause 6.2*
- (c) Additional point at no 07 of Clause 7.1 for inspection & Testing*
- (d) Modification in clause 7.1 in line with IS/ASTM standards*
- (e) Responsibility for implementation changed to HOD(EM) in clause 8.0*
- (f) Annexure added at Clause 11.0 containing list of tools, and checklist for inspection and testing.*

6.0 THE SPECIFIC INSTRUCTIONS TO BE FOLLOWED:

All users of these hand held tools for live working shall follow the contents of this LMI without fail.

6.1 PURCHASE OF INSULATING HAND TOOLS

Insulated hand tools which meet the appropriate standard for the insulation and the appropriate standard for the type of tool should meet the requirements for most application of live working up to 1000 V A.C.(*Applicable standards IS13772:2017 or ASTM F1505-16*)<Rev 2> . If there is an additional requirement this can be included in the purchase specification and it is insisted for CPRI test certificate. In order to prevent confusion between “insulating hand tools” and general engineering hand tools with handles made of insulating materials it is recommended that only one colour of insulating hand tool be used on a site and that other general engineering tools do not use that colour.

6.2 CONTROL AND ISSUE OF INSULATED HAND TOOLS

A policy should be prepared containing the following information to regulate the practice of how insulated hand held tools are issued to staff <Rev 2>.

1. The tools have to be readily available so that staff do not take short cuts and use unsuitable tools.
2. The tools need to be regularly inspected and tested.
3. Marking of the tools or fixing of labels in order to identify each item for the purpose of keeping track can impair its performance
4. Do not use insulating tools as general purpose engineering tools can lead to rapid deterioration of the insulating systems due to mechanical damage and contamination by oil, dirt and moisture.
5. Issue the insulated hand tools as kits in special tool cases or bags. The cases or bags should be then numbered for the purpose of control and testing. The kits can be on personnel issue or issued when the job requires it.

6.3 STORAGE

Insulated tools should be stored in clean, dry conditions and in such a manner that the risk of damage to the insulation is minimized. They should also be protected against direct sunlight.

7.0 METHODOLOGY OF IMPLEMENTATION:

1. Insulated tools must always be used in such a manner as to preserve the integrity of the insulation and thus ensure continued safety in use.
2. Insulated tools must be kept clean and dry during use. Accidental contamination of the insulation should be avoided by prompt removal of oil, compound splashes, etc., followed by thorough drying of the tool. Unnecessary exposure to direct sunlight for long periods should be avoided.

3. Insulated tools should only be used for their designed purpose and the correct tool should always be selected for the intended application.
4. Under no circumstances must any insulation be removed, or a tool otherwise modified.

7.1 INSPECTION AND TESTING

All insulated tools should be inspected before use for evidence of mechanical damage or physical deterioration which might impair electrical performance. If there is any doubt about the quality of the insulation, the tool should be withdrawn from service and replaced. No attempt should be made to repair the tool. Tool should be inspected before every use. All tools should be regularly inspected and tested, and any found to not meet the requirements should be withdrawn from service.

The inspection process should include the following features.

1. Check for mechanical cuts and abrasion to the insulation
2. Check for burns or solvent attack.
3. Check for cracks
4. Check that the bonding between the insulation and the metal parts is intact. Look for evidence of the insulation moving relative to the metal parts and the insulation parting from the metal parts at the point where the insulation finishes.
5. Check for general wear damaging the surface finish so that the insulation surface properties will be susceptible to moisture, oil and dirt.
6. Check that the insulation is clean and free from oil, grease and dirt
7. *The identified tools must be re-tested after 1 year or if found unsuitable during any inspection. Re-testing to be done in line with IS/ASTM standards. The test and inspection records are to be maintained.<Rev 2>*

Following the inspection the tools should be subjected to 10KV, 50 Hz for 3 minutes between the metal part and outer surface of

the insulation. To achieve this the tool should be immersed in water with a conductivity not less than 250 micro siemens percm at ambient temperature. The tool should be immersed to a depth such that 24mm + 2mm of insulation adjacent to the exposed metal parts is above the water level. Where this cannot be achieved metal foil such as aluminum cooking foil can be wrapped around the insulation in a manner that the foil does not reach within 24mm of any exposed metal part of the tool. The test is considered successful if no breakdown is detected.

8.0 THE RESPONSIBILITY CENTERS FOR IMPLEMENTATION:

HOD(EM)<Rev 2>

9.0 THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:

Any deviation shall be reported to HOD (EM).

10.0 REVIEW:

Document will be reviewed once in two years by HOP (Simhadri).

11.0 Annexure

Annexure A1

List of insulated tools

- i) Insulated screw driver*
- ii) Insulated wrench*
- iii) Insulated combination stripper*
- iv) Insulated nut driver*
- v) Insulated pliers and nose pliers*
- vi) Insulated crimping tools*
- vii) Insulated earthing rods*

Annexure A2

Checklist

<i>Sl no</i>	<i>Description of inspection</i>	<i>ok</i>	<i>Not ok</i>
1	<i>Prior to use, physically inspect insulated tools</i>		
2	<i>Store insulated tools in clean and dry condition</i>		
3	<i>Keep insulated tools clean and after use place them into the designated tool cases or bags.</i>		
4	<i>Periodically inspect insulated tools. Keep inspection record readily available. Periodicity of inspection-yearly(but if the tools are used in an environment which can lead to rapid degeneration the interval between inspection should be reduced)</i>		
5	<i>Check for mechanical cuts and abrasion to the insulation</i>		
6	<i>Check for burns or solvent attack.</i>		
7	<i>Check for cracks.</i>		
8	<i>Check that the bonding between the insulation and metal parts is intact.</i>		
9	<i>Check for general wear damaging the surface finish so that the insulation surface properties will be susceptible to moisture, oil and dirt</i>		
10	<i>Check that the insulation is clean and free from oil, grease and dirt</i>		
11	<i>Ensure that tools must be re-tested after 1 year or if found unsuitable during any inspection</i>		

Annexure A3

Sl no	Testing procedure	Result ok/not ok
1	<i>Following the annual inspection, the tools should be subjected to 10KV, 50 Hz for 3 minutes between the metal part and outer surface of the insulation. To achieve this the tool should be immersed in water with a conductivity not less than 250 micro siemens per cm at ambient temperature. The tool should be immersed to a depth such that 24mm + 2mm of insulation adjacent to the exposed metal parts is above the water level. Where this cannot be achieved metal foil such as aluminum cooking foil can be wrapped around the insulation in a manner that the foil does not reach within 24mm of any exposed metal part of the tool. The test is considered successful if no breakdown is detected.</i>	